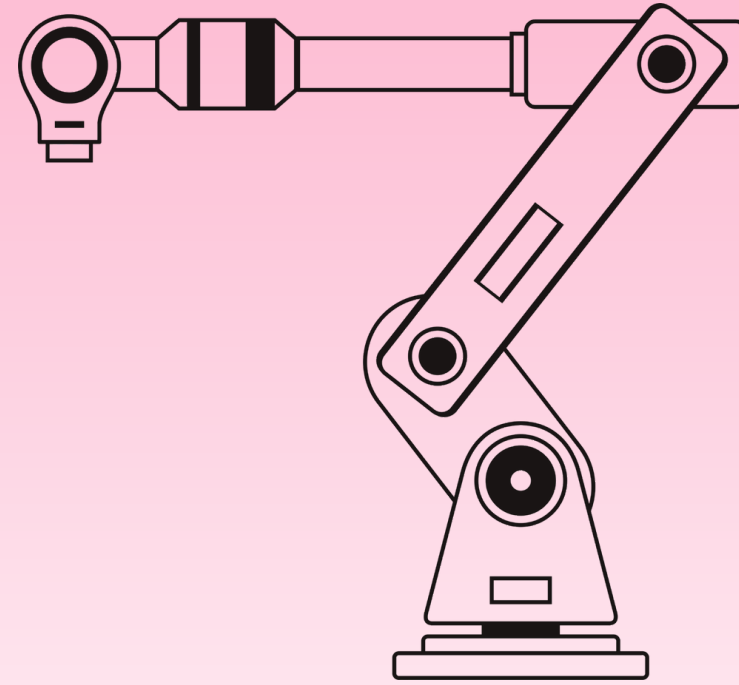
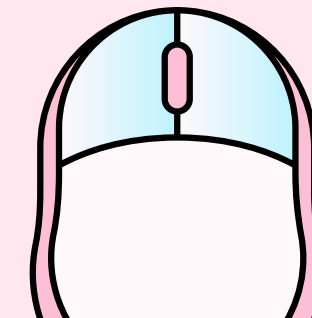
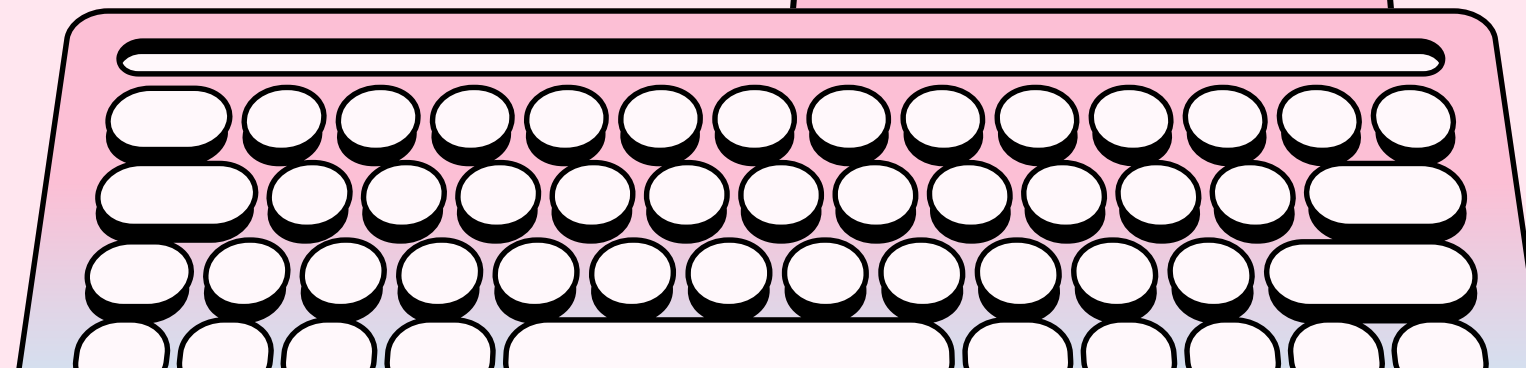
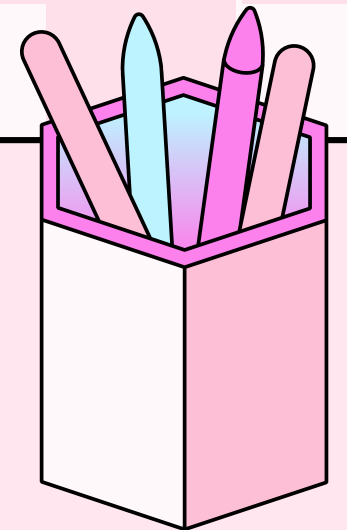
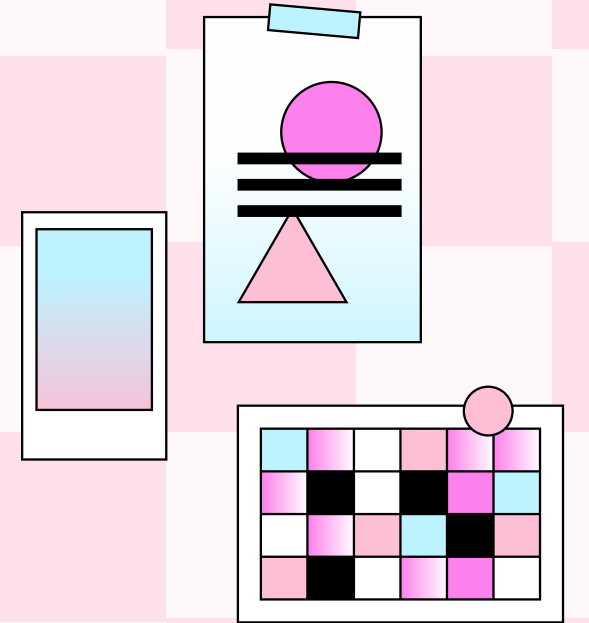
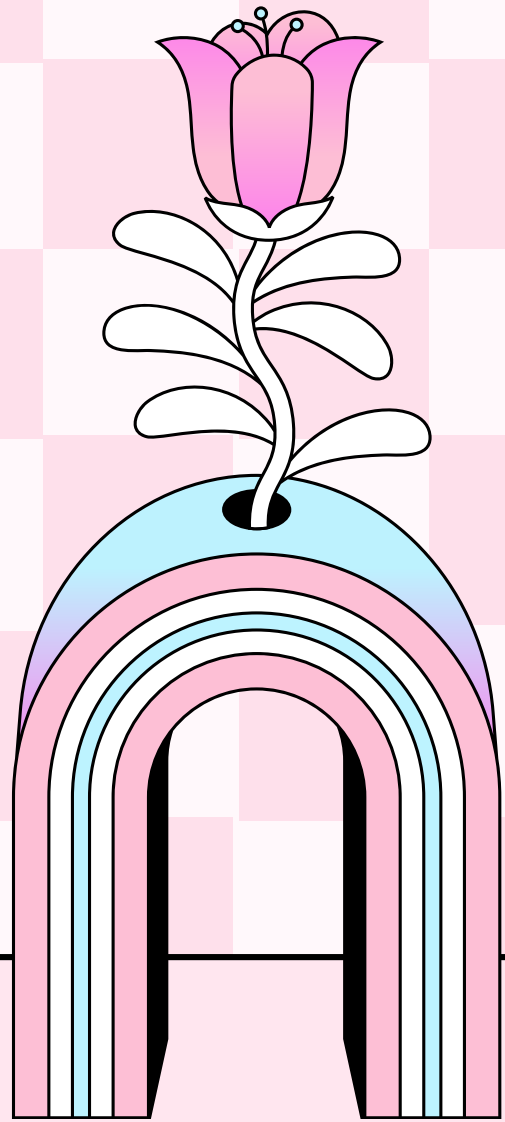
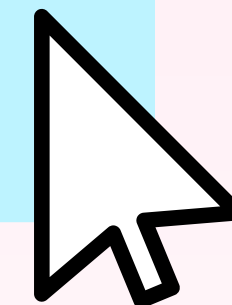




## 2.1.2 A Robotic Revolution

HARSHITA





RESEARCH

# RESEARCH QUESTIONS

- WHAT IS A ROBOT, AND HOW IS IT DEFINED?
- WHO CREATED THE EARLIEST ROBOTS?
- WHAT INVENTIONS LED TO MODERN ROBOTICS?
- HOW HAVE ROBOTS CHANGED INDUSTRIES?
- HOW MIGHT ROBOTS EVOLVE IN THE NEXT 50 YEARS?

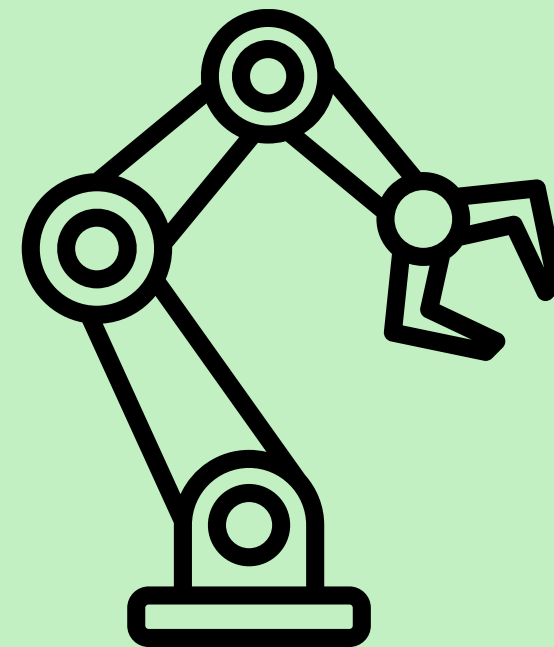


# TIMELINE: 5 MAJOR EVENTS IN ROBOTICS HISTORY

- **1921 — THE WORD “ROBOT” IS INTRODUCED**
  - KAREL ČAPEK’S PLAY R.U.R. BRINGS THE TERM “ROBOT” INTO THE WORLD.
- **1954 — FIRST PROGRAMMABLE ROBOT (UNIMATE)**
  - GEORGE DEVOL CREATES THE FIRST INDUSTRIAL ROBOT ARM.
- **1966 — SHAKY THE ROBOT**
  - THE FIRST MOBILE ROBOT WITH BASIC ARTIFICIAL INTELLIGENCE.
- **1997 — SOJOURNER MARS ROVER**
  - THE FIRST ROBOTIC ROVER TO EXPLORE ANOTHER PLANET.
- **2002 — ROOMBA ROBOT VACUUM RELEASED**
  - ROBOTICS ENTERS EVERYDAY HOME LIFE FOR THE FIRST TIME

# HOW ROBOTICS HAS CHANGED THE WORLD

- FASTER AND SAFER MANUFACTURING
- IMPROVED MEDICAL SURGERIES
- SPACE EXPLORATION MADE POSSIBLE
- EVERYDAY HOME AUTOMATION
- ASSISTIVE TECHNOLOGY FOR DISABILITIES





# FUTURE ROBOTICS: 3 EVENTS THAT WILL SHAPE THE NEXT 50 YEARS

- **FULLY AUTONOMOUS TRANSPORTATION SYSTEMS**

- SELF-DRIVING DELIVERY VEHICLES, BUSES, AND ROBOTIC LOGISTICS WILL RESHAPE CITIES.

- **ADVANCED MEDICAL CARE ROBOTS**

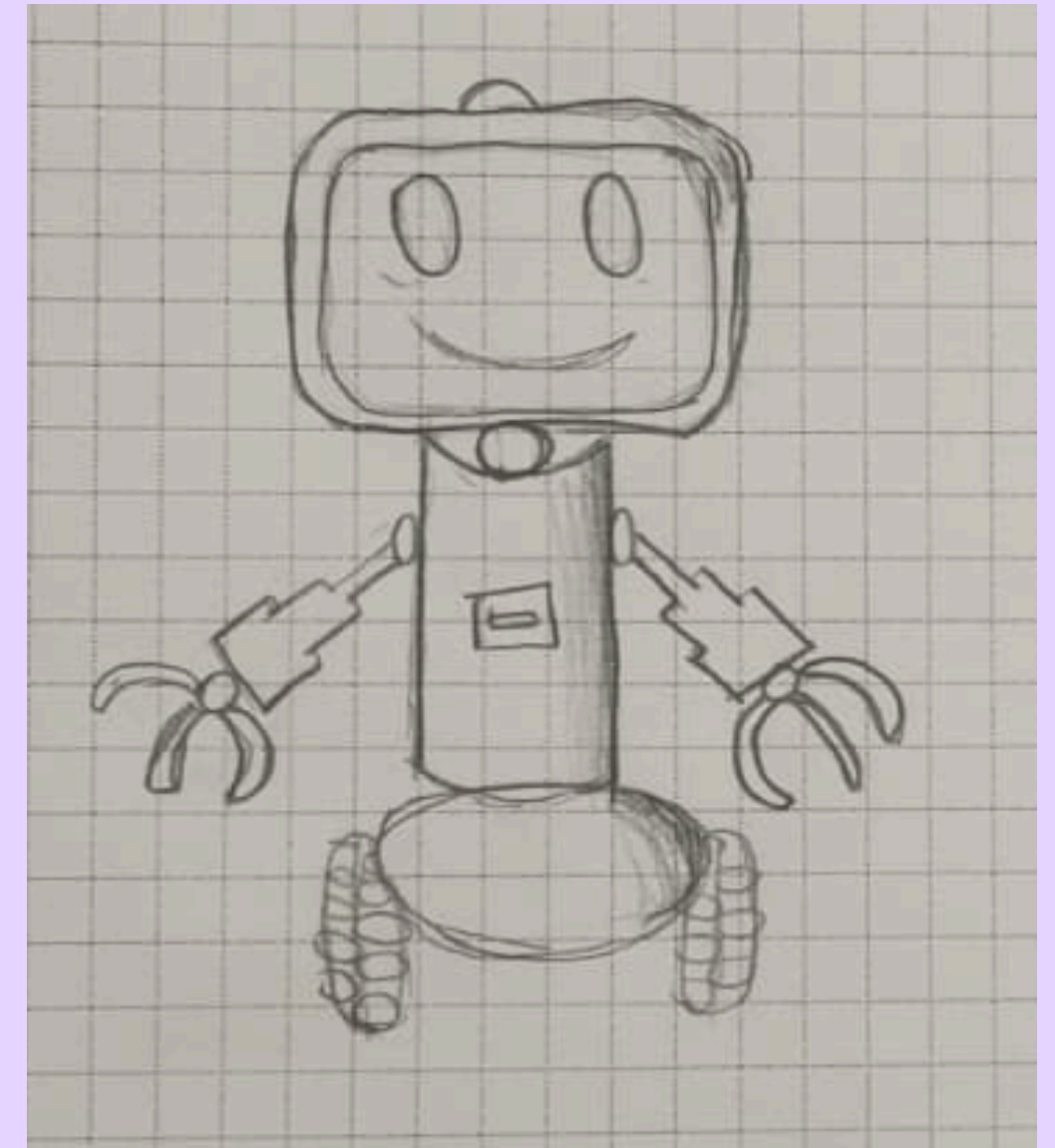
- ROBOTS WILL HELP ELDERLY PEOPLE, ASSIST IN PHYSICAL THERAPY, AND SUPPORT DOCTORS WITH PRECISE SURGERIES.

- **AUTOMATED CONSTRUCTION & DISASTER RESPONSE ROBOTS**

- ROBOTS WILL BUILD HOUSES, REPAIR INFRASTRUCTURE, AND HELP RESPOND TO FIRES, STORMS, AND DANGEROUS ENVIRONMENTS.

# MY FUTURE ROBOT CONCEPT: "SYNTHIA"

- **PURPOSE:** A HOME ORGANIZATION + ASSISTANT ROBOT
- **JOBS:** PUT ITEMS AWAY, SORT LAUNDRY, REMIND USERS OF SCHEDULES, HELP CLEAN
- **MOBILITY:** TWO LARGE OMNI-WHEELS
- **ARMS:** TWO TELESCOPING ARMS WITH SOFT, RUBBERIZED THREE-FINGER GRIPS
- **HEAD:** LED SCREEN FACE WITH SENSORS
- **HEIGHT:** ABOUT 4 FT TALL
- **EXTRAS:** STORAGE COMPARTMENT INSIDE THE TORS



# CONCLUSION

- **HOW DO I “CLASSIFY” WHAT A ROBOT IS?**
  - A ROBOT IS A PROGRAMMABLE MACHINE DESIGNED TO DO TASKS ON ITS OWN OR WITH LIMITED HUMAN HELP.
  - ROBOTS USE SENSORS TO GATHER INFORMATION, PROCESSORS TO MAKE DECISIONS, AND MOTORS OR ACTUATORS TO MOVE OR COMPLETE ACTIONS
- ROBOTICS BEGAN WITH **SIMPLE** MECHANICAL IDEAS AND NOW IMPACTS NEARLY EVERY PART OF LIFE.
- WITH **RAPID ADVANCEMENTS** IN AI, MOBILITY, AND AUTOMATION, THE NEXT 50 YEARS WILL BRING EVEN MORE POWERFUL AND HELPFUL ROBOTS THAT SHAPE CITIES, HEALTHCARE, AND EVERYDAY LIVING.